Quantifying Atmospheric Density and Orbital Error Distributions During Geomagetic Storm-Time

Final Project
STAT 5630: Computational Bayesian Statistics
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1. Introduction

This report presents two complementary studies focused on atmospheric density uncertainty and its impact on orbital prediction:

1. **Exp1**: A Monte Carlo analysis examining how atmospheric mass density distributions propagate into orbital position error distributions.
2. **Exp2**: A Bayesian regression of atmospheric mass density using Hamiltonian Monte Carlo (HMC), incorporating spatial and physical drivers of variability.

Together, these experiments address both forward uncertainty propagation (Exp1) and statistical modeling of atmospheric density (Exp2). This Introduction section outlines the background, motivation, and research objectives for both experiments.

1.1 Background

Low Earth Orbit (LEO) is a highly dynamic environment where atmospheric neutral mass density exhibits strong spatial and temporal variability driven by solar radiation, geomagnetic activity, and other physical dynamics. Satellites operating in LEO (typically 400-1000 km altitude) experience drag acceleration due to the density. Because the density cannot be continuously observed in all space and time, it is represented through empirical or physics-based models that introduce significant uncertainty. This modeling uncertainty is a dominant contributor to orbit prediction error, particularly over long propagation intervals.

The solar wind is a continuous plasma flow of charged particles (mainly electrons and protons) emitted by the Sun, carrying the interplanetary magnetic field. Geomagnetic storms arise from enhanced solar wind, typically due to coronal mass ejections (CMEs), coupling with Earth's magnetosphere. These storms deposit energy into the thermosphere, increasing temperature and causing atmospheric expansion. Geomagnetic storms are known to significantly increase density and its variability, amplifying the drag experienced by satellites [2]. The resulting storm-time dynamics are highly nonlinear [7] and may induce non-Gaussian density uncertainties. As a result, orbital errors during storms are not only larger in magnitude but also less predictable than under quiet-time conditions [1].

Geomagnetic and solar activity are commonly quantified using standard indices. The Dst index measures the strength of the ring current in nT and provides a global indicator of geomagnetic storm intensity, with more negative values corresponding to stronger storms. The Ap index is a planetary geomagnetic index derived from ground-based magnetometer measurements and reflects short-term geomagnetic activity on a quasi-linear scale. The f10.7 index represents solar radio flux at a wavelength of 10.7 cm and serves as a proxy for solar EUV radiation. Together, these indices provide key inputs for characterizing and solar and geomagnetic activity.

The High Accuracy Satellite Drag Model (HASDM) is a data-assimilative thermospheric density model that combines physics-based atmospheric modeling with satellite drag observations to produce near-real-time estimates of global mass density in LEO. It is widely used in space operations because it provides higher accuracy than purely empirical models by continuously correcting the background atmosphere using tracking data from multiple satellites. This model will be used for density data in this report since it blends observed data with physical laws.

Current orbit tracking frameworks rely on simplified atmospheric models and typically assume Gaussian-distributed state errors characterized entirely by covariance matrices for computational efficiency [4]. However, during geomagnetic storms, both the mean and variance of density increase substantially, and the distribution exhibits measurable departures from Gaussianity, as shown by Peters et al. [6]. Ensemble simulations of atmospheric states (100 particle filter realizations from TIE-GCM model) confirm the emergence of heavier tails and skewed behavior under storm-time compared to quiet-time. Representative quiet and storm-time density profiles along an orbit are shown in Figure 1, where storm conditions exhibit both elevated peaks and increased variance.

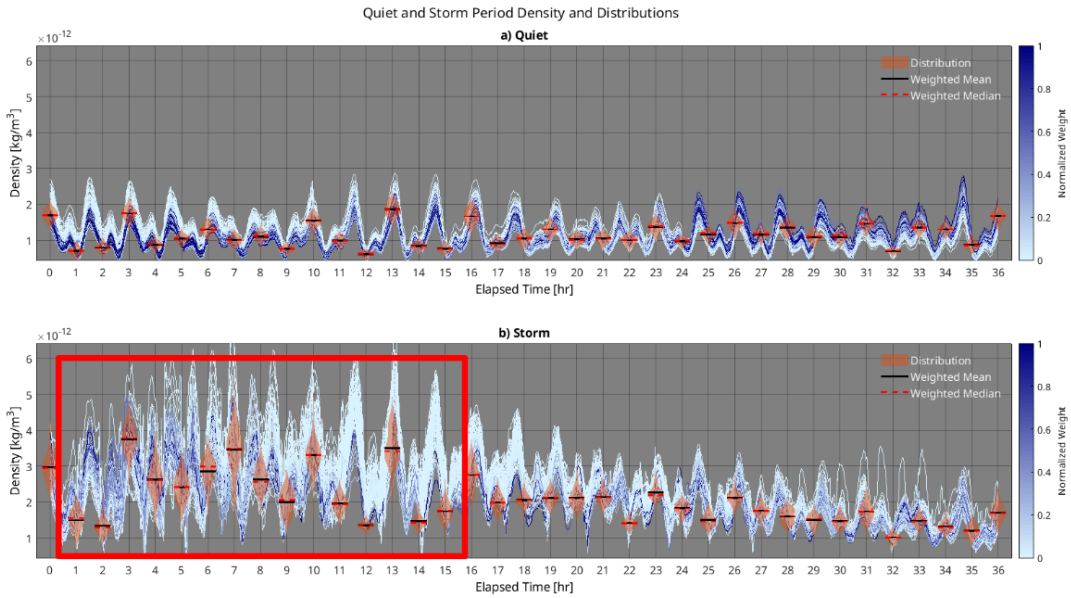

Figure 1: Mass density along a simulated spacecraft trajectory under quiet (top) and storm-time (bottom) conditions. The boxed region indicates enhanced density during the storm peak [6].

These density changes directly propagate into drag acceleration uncertainty. Tracking statistical moments of density and propagated orbital position errors reveals that statistics (mean, variance, and higher-order moments) are transmitted through the dynamics, producing corresponding non-Gaussian state uncertainty. This is shown in Figure 2 where the storm-time statistics are shown in blue and the quiet-time statistics in black.

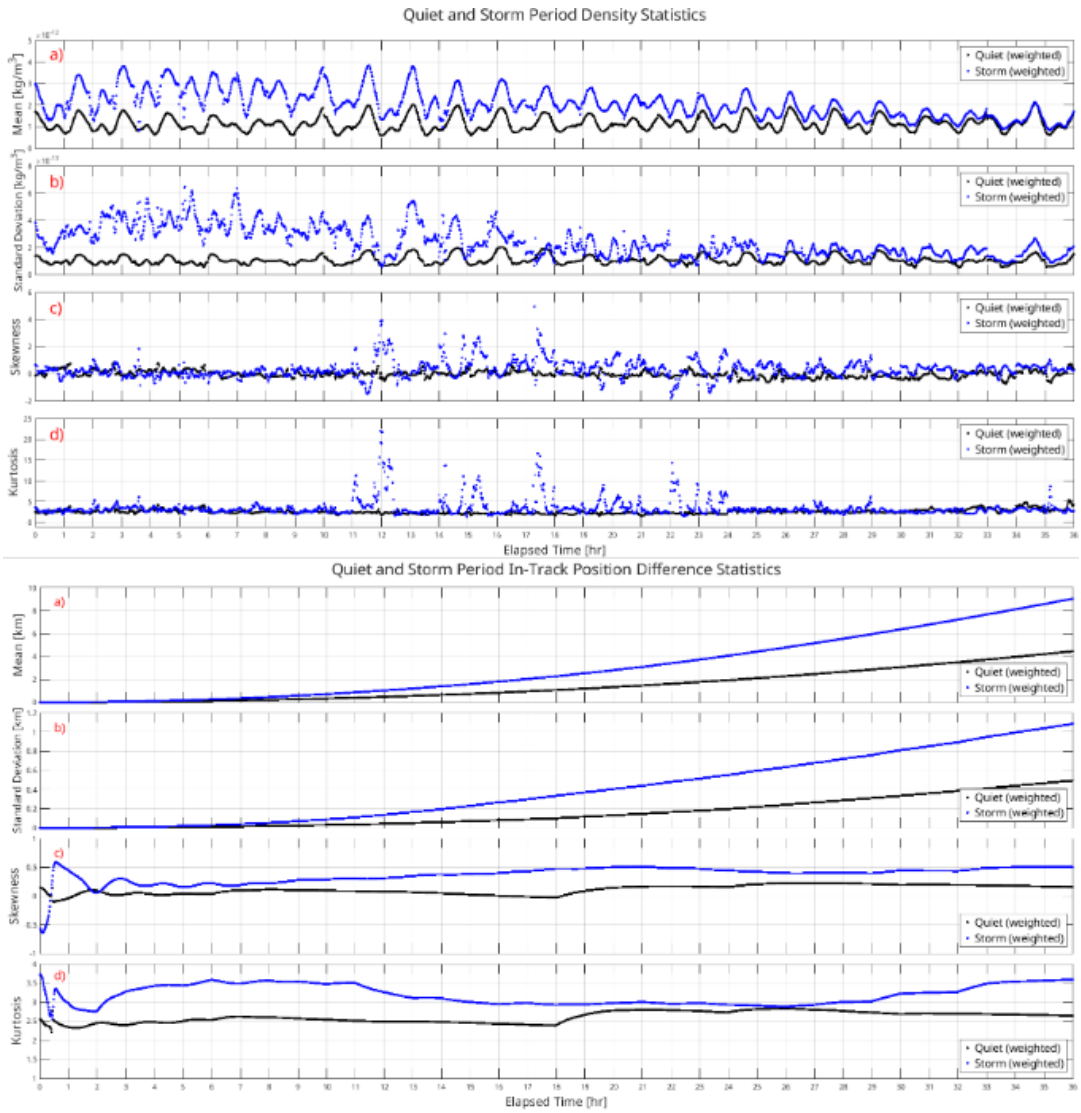

Figure 2: Evolution of statistical moments of atmospheric density (top) and orbital error (bottom) over time [6].

1.2 Motivation

Satellites re-enter Earth's atmosphere at end-of-life, upon malfunction, or when atmospheric drag leads to orbital decay beyond correction capabilities. Between 2020 and 2024, 523 Starlink satellites re-entered [5]. Oliveria et al. showed a clear dependence between reentry rates and geomagnetic activity [5]. Orbital decay rate and associated prediction error exhibited an approximately linear dependence on geomagnetic activity ($R \approx 0.85$), reinforcing the need for improved atmospheric density and uncertainty modeling during storm-time.

Existing atmospheric density models present a fundamental trade-off between accuracy and computational efficiency. Empirical models are typically fast and suitable for real-time orbit propagation, but they become insufficiently accurate under geomagnetic storm conditions. In contrast, physics-based models can capture atmospheric behavior more accurately, but they are often too computationally expensive for operational real-time orbit prediction.

Despite extensive use of both approaches, the relationship between atmospheric density variability and resulting orbital prediction errors remains insufficiently quantified. This gap limits the ability to systematically assess uncertainty growth in orbit propagation under varying space weather conditions.

Previous work has investigated the propagation of atmospheric density model uncertainty into orbital uncertainty. In particular, Doornbos et al. studied how grid-scale density model uncertainties translate into uncertainties in orbital states using Monte Carlo simulations and analytical comparison methods [3]. Their results demonstrated how

density uncertainty can be mapped into corresponding orbit prediction uncertainty, providing a basis for uncertainty propagation in orbit determination.

However, their approach was limited to characterizing uncertainty using only the standard deviation with Gaussian-distributed density perturbations. As a result, higher-order statistical effects and non-Gaussian behavior in density variability were not captured.

1.3 Exp1: Research Goal and Objectives

The goal of this research experiment is to improve uncertainty quantification (UQ) of orbital position errors during geomagnetic storm conditions, with the aim of enabling more reliable tracking and decision-making for space traffic operators.

The first objective is to examine the impact of non-Gaussian atmospheric density on orbital errors. This includes propagating non-Gaussian density distributions through orbital dynamics and comparing the resulting distributions to ones produced by Gaussian density. The second objective is to quantify the effect of increased density variance on orbital position uncertainty. This involves comparing orbital error variance generated using high-variance density distributions versus low-variance cases. The final objective is to separate the effect of increased variance from higher-order statistical moments such as skewness and kurtosis.

Figure 3 explains that the link between atmospheric density and orbital position errors is more straight forward than the link between physical conditions to atmospheric density.

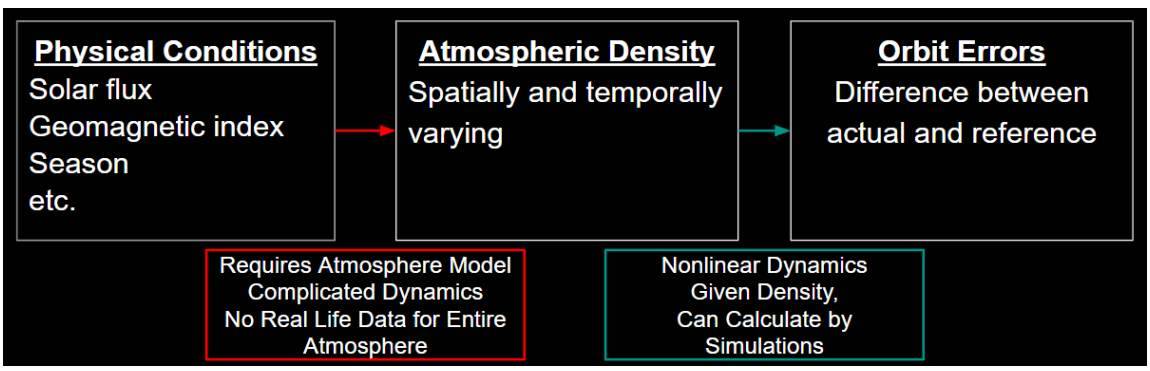

Figure 3: Various factors that need to be accounted for to establish links between physical conditions, atmospheric density, and orbital errors.

1.4 Exp2: Research Goal and Objectives

The goal of this experiment is to identify and quantify the physical and spatial drivers of atmospheric mass density. This is achieved by performing a Bayesian regression using Hamiltonian Monte Carlo, allowing joint inference of parameter uncertainty and their contributions to density variability.

2. Exp1: Methods

This experiment constructs four marginal probability distributions for atmospheric density. Each distribution is used to generate density realizations, which are then propagated through orbital dynamics via Monte Carlo simulation. The resulting ensemble of trajectories is used to compute orbital position errors, yielding a probability distribution of orbital position error for each density case. Figure 4 displays a visual of these steps.

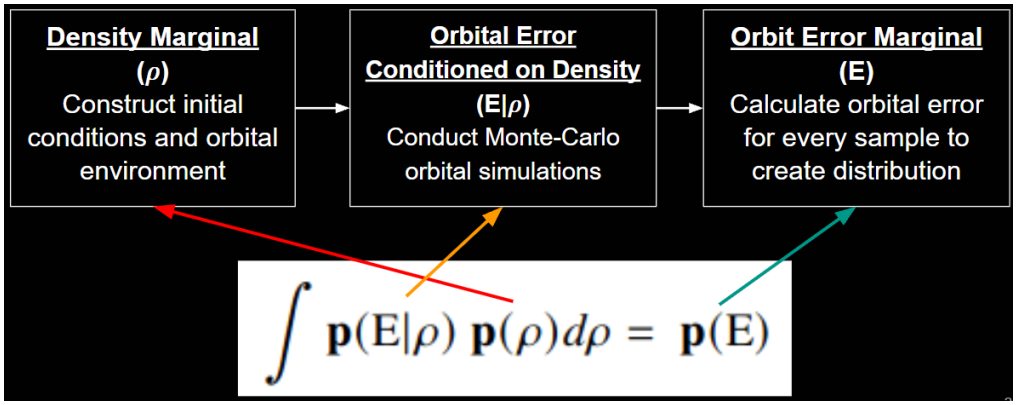

Figure 4: Method for Exp1.

2.1 Orbital Error Definition

Orbital position error is analyzed in the Radial–Intrack–Crosstrack (RIC) coordinate frame. RIC coordinates are commonly used for orbital error analysis. In this frame, the in-track (I) direction is aligned with the spacecraft's velocity vector and is therefore the primary direction influenced by atmospheric drag. The in-track position error is defined relative to the mean orbit and accumulates over time due to continuous drag acceleration. Figure 5 displays the RIC axes.

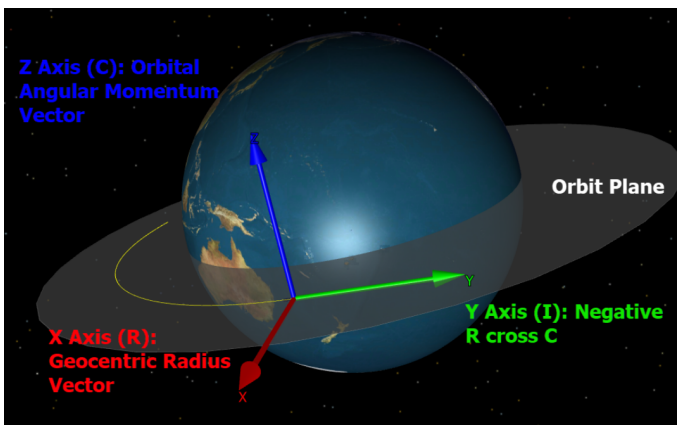

Figure 5: RIC coordinate frame.

2.2 Dataset

Atmospheric mass density data are obtained from HASDM for the Gannon geomagnetic storm period (May 9–13, 2024). The HASDM grid is coarse in both space and time resolution. To improve the experiment's resolution, a refined spatiotemporal density grid is constructed by interpolating the atmospheric mass density in space and time. The interpolation is performed in log space to better preserve relative density variations. To reduce computational and memory requirements, the domain is further truncated by excluding high-latitude regions above 20° and below −20°. This restriction limits the grid size while retaining a primary region of interest. Figure 6 shows the two grids and their resolutions.

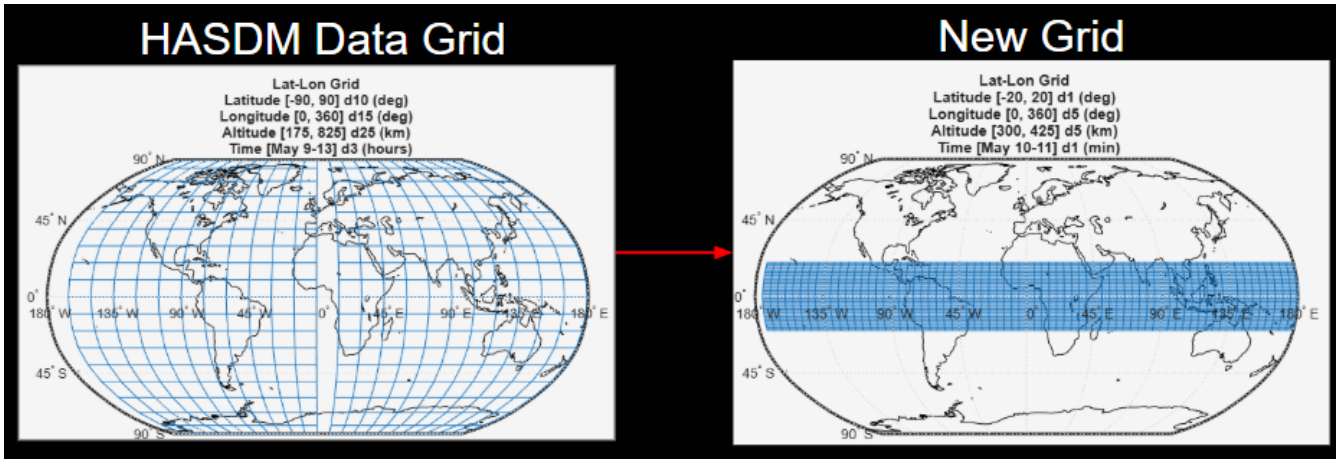

Figure 6: HASDM data grid and new grid used.

The new full atmospheric density grid has very high dimensionality, with a resolution of

$$41 \times 70 \times 26 \times 2881 = 214{,}980{,}220$$

grid points. A single density field at this resolution requires approximately 1.45 GB of memory, making repeated sampling and propagation computationally expensive. Additionally, no sub-grid interpolation is used in order to avoid further increases in computational cost and complexity.

To reduce the dimensional burden, the density field is reduced to a one-dimensional statistical representation by introducing a scalar scaling factor α applied to the mean density field. In this formulation, density realizations are generated as scaled perturbations of the HASDM data, rather than as fully spatially and temporally varying fields. This approach sacrifices detailed local variability in exchange for computational efficiency. This means that it is assumed that the atmospheric density is evenly scaled regardless of location and time. The drag acceleration vector becomes

$$\mathbf{a_{drag}} = -\rho_i(\mathbf{r}, t)\frac{C_B}{2}v\mathbf{v} \rightarrow -\alpha_i\rho_{\text{mean}}(\mathbf{r}, t)\frac{C_B}{2}v\mathbf{v}$$

where ρ is the density, C_B is a constant based on spacecraft parameters, $\mathbf{v} = v\hat{\mathbf{v}}$ is the relative velocity vector of the satellite with respect to the atmosphere, and $(\mathbf{r}, t)$ represents the spacecraft's location and time.

2.3 Density Distributions

The refined HASDM density field is used as the mean atmospheric state. Density variability is modeled using a reduced-order formulation based on the scaling factor α, such that each realization is generated by scaling the mean field. Statistical moments for the distributions are prescribed from prior analysis [6] and Figure 2, with coefficient of variation $C_v = \sigma/\mu \approx 0.1$, skewness $= 1$, and kurtosis ≈ 10. A total of $N = 200$ samples are drawn from each distribution.

To isolate the influence of different statistical properties, four distributions for α are considered. Gaussian cases consist of a (1) small-variance distribution and a (2) large-variance distribution, allowing the effect of increased variance to be isolated. Non-Gaussian cases include a (3) skewed distribution (implemented via a skew-normal distribution with shape parameter $a = 10$) and a (4) heavy-tailed distribution (Student-t with $\nu = 3$) to represent elevated kurtosis and strongly non-Gaussian behavior.

Together, these four cases enable separation of the effects of variance, skewness, and kurtosis on density-driven orbital errors. The four distributions of α are shown in Figure 7. All distributions are normalized to have mean approximately equal to 1, ensuring that the mean density corresponds to the unscaled HASDM field. The coefficient of variation is approximately $C_v \approx 0.1$ for all cases except the large-variance Gaussian, where $C_v = 0.20$. The Gaussian distributions exhibit only negligible skewness and relatively low kurtosis compared to the heavy-tailed case; however, since both Gaussian cases share similar higher-order behavior, direct comparison between them remains valid for isolating variance effects.

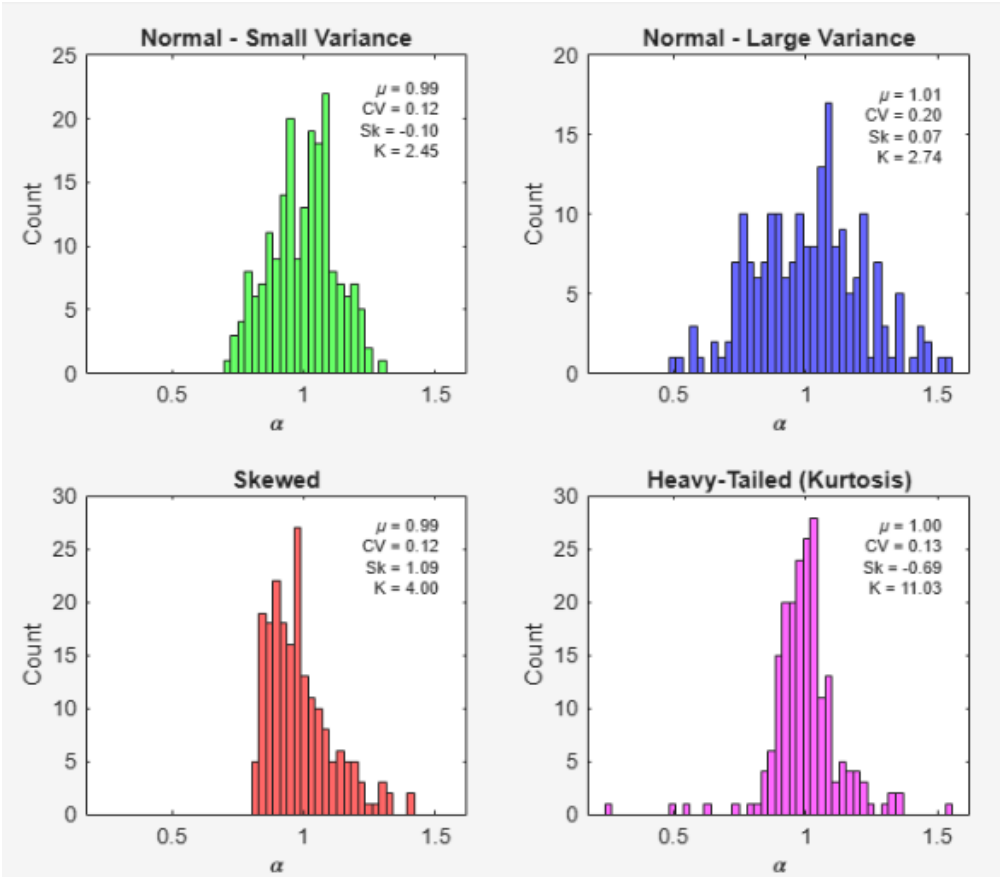

Figure 7: Probability distributions of the density scaling factor α used in the analysis.

2.4 Orbit Propagation Simulation Setup

A Monte Carlo framework is used to propagate orbital states under sampled atmospheric density realizations. For each distribution, the orbit is propagated through the 200 α samples, producing 200 distinct trajectory realizations. The simulation spans May 10–13 with a 10-minute output timestep. The computational cost is approximately 8.6 seconds per sample.

To isolate the effect of density variability, all simulations are initialized from identical orbital states. The chosen orbit is a common circular LEO with 400 km altitude and 45° inclination. Typical values for the satellite's area, mass, and drag coefficient ($C_D = 2.2$) are used. The orbital dynamics are point-mass gravity, Earth's oblateness (J2) perturbation, and atmospheric drag acceleration. Higher-order perturbations are neglected since it is assumed that their effects are negligible.

3. Exp1: Results

3.1 Validation of Propagation

A validation run is performed using the mean density field ($\alpha = 1$) to verify correct implementation of the propagation framework. Figure 8 shows the resulting trajectory in 3D, along with latitude, longitude, and altitude over time. The satellite remains within the bounds of the refined HASDM grid throughout the simulation, indicating no extrapolation errors.

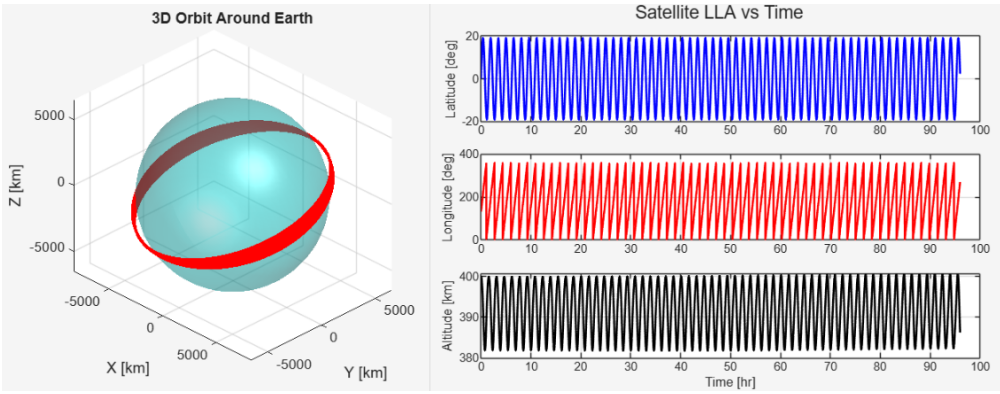

Figure 8: Satellite orbit in 3D (left) and latitude, longitude, and altitude (right).

Figure 9 shows the atmospheric density along the satellite trajectory. Elevated density values are observed during the peak of the geomagnetic storm (red box). The refined grid extends through May 11, while propagation continues to May 13; thus, density fields for May 12–13 are constructed by repeating the May 11 data. This enables observation of cumulative orbital error over a longer time while limiting memory requirements. The repeated structure is visible in the green box. Periodic oscillations in density correspond to the satellite's orbital motion, reflecting higher density on the dayside and lower density on the nightside. This is a known effect.

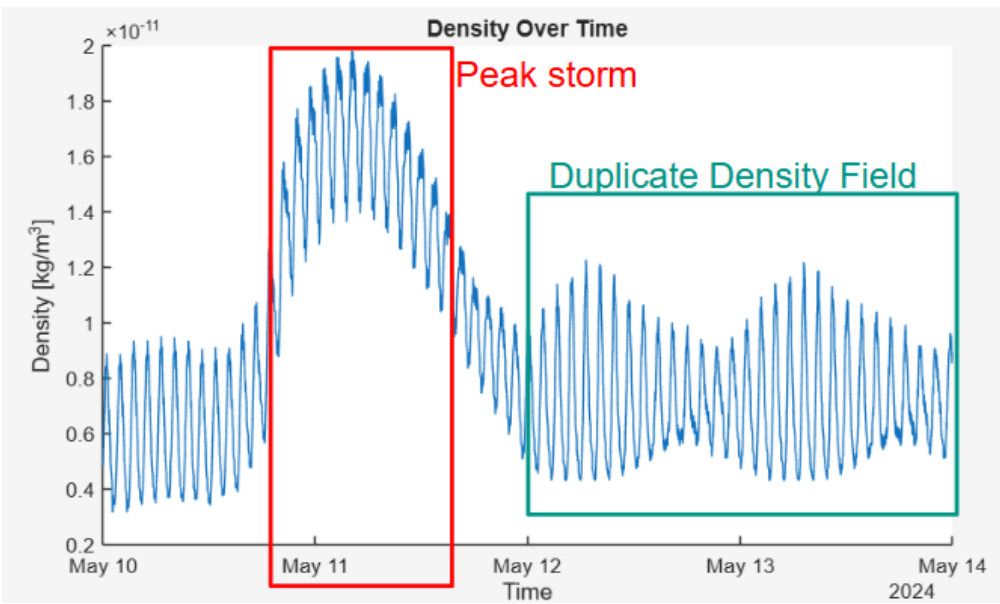

Figure 9: Atmospheric density along the satellite trajectory over time.

3.2 Orbital Errors

Orbital error distributions are characterized by computing statistical moments across the 200 samples at each time step for each density case, with the same method as Peters et al. [6]. Figure 10 shows the time evolution of these moments. The coefficient of variation, skewness, and kurtosis remain approximately constant over time, while the mean error grows due to the cumulative nature of drag-induced perturbations.

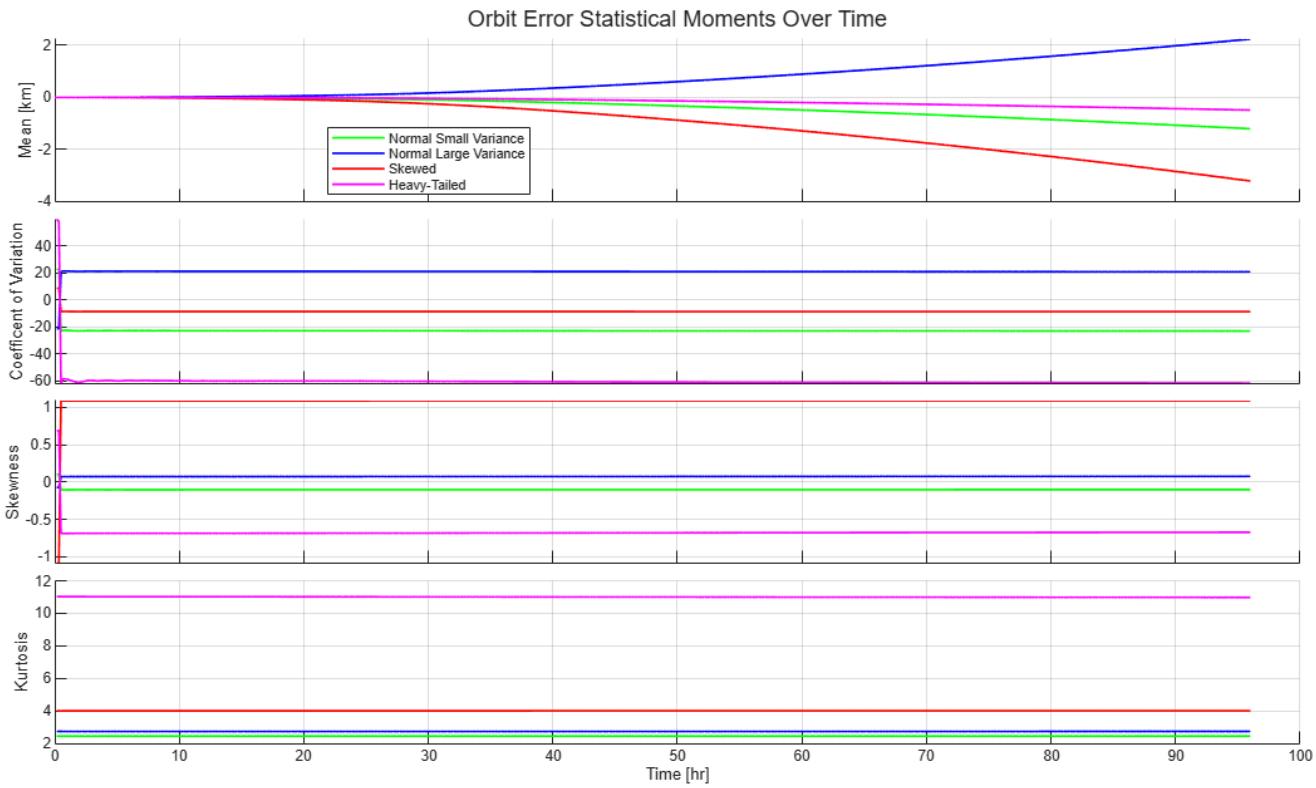

Figure 10: Time evolution of statistical moments of in-track orbital error for each density distribution.

To compare distributions directly, the orbital error distributions at the last time step are plotted in Figure 11. The orbital error distributions closely match their corresponding α distributions (Figure 7), indicating that the statistical structure of atmospheric density is largely preserved through orbit propagation.

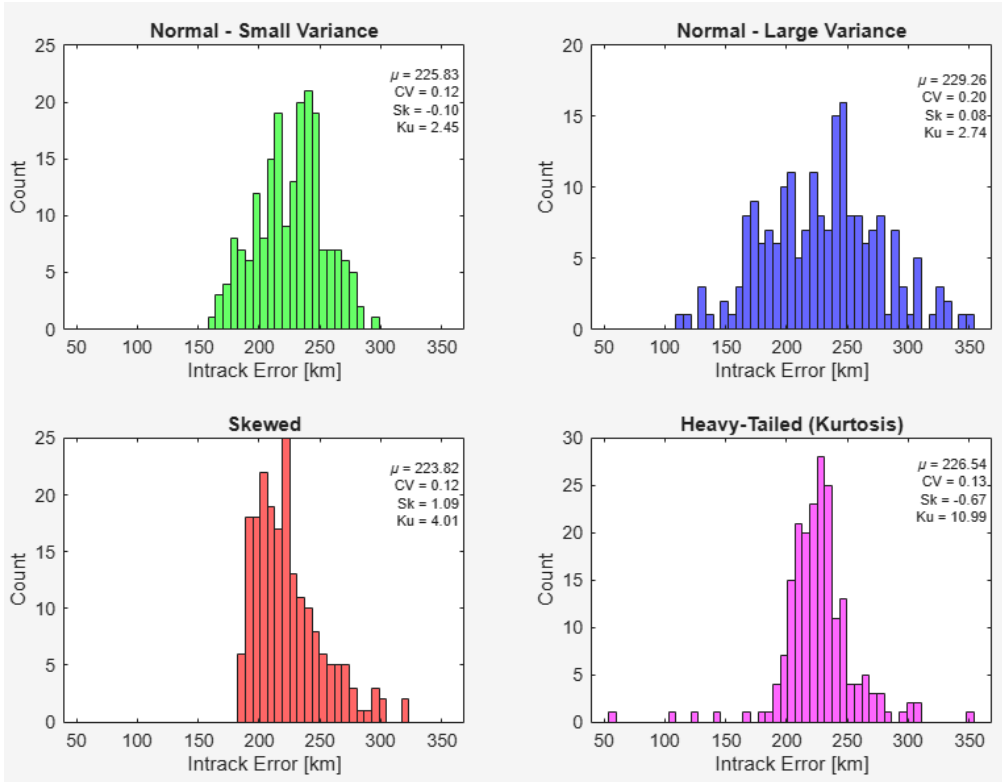

Figure 11: Orbital error distributions at the final time step.

3.3 Exp1: Conclusions

This experiment addresses the impact of non-Gaussian atmospheric density variability on orbital position uncertainty during geomagnetic storm conditions. The results show that non-Gaussian features in the density distributions are preserved through orbit propagation. The coefficient of variation, skewness, and kurtosis of the resulting orbital error distributions remain approximately constant over time and closely match those of the corresponding α distributions. This demonstrates that non-Gaussian density behavior directly maps into orbit uncertainty, satisfying the first objective.

The effect of increased density mean is clearly observed in the growth of orbital error. Density distributions with a higher mean produce larger mean orbital errors, while maintaining consistent normalized variability (coefficient of variation). This isolates mean as the primary driver of error magnitude while the variance directly maps through orbit propagation, addressing the second objective.

By comparing Gaussian and non-Gaussian cases, the influence of higher-order moments is separated from variance effects. While variance controls the spread of error growth, skewness and kurtosis determine the shape of the orbital error distribution. These higher-order characteristics persist independently of variance, confirming that they contribute distinct and measurable effects to orbit uncertainty, satisfying the third objective.

Overall, the results establish a direct and structured mapping between atmospheric density statistics and orbital error distributions. Given a density distribution and mean error growth, the full orbital error distribution can be inferred. This provides a foundation for improved uncertainty quantification in orbit prediction, particularly under geomagnetic storm conditions, and supports more reliable space traffic management.

3.4 Future Work

Future work will focus on improving both statistical accuracy and computational scalability. Increasing the Monte Carlo sample size to $N = 1000$ will improve convergence of higher-order moments, reduce sampling noise, and ensure closer agreement in distribution means across cases. Additional improvements include comparing different orbits and altitudes and incorporating linear sub-grid interpolation in space and time to better resolve fine-scale density

variations. However, this will significantly increase computational cost.

4. Exp2: Methods

This experiment performs a linear regression of atmospheric mass density using HMC. The model is defined as

$$\log(\text{Density}) = \beta_0 + \beta_1\,\text{LAT} + \beta_2\,\text{LST} + \beta_3\,\text{Dst} + \beta_4\,\text{ap} + \beta_5\,\text{f10.7} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \tau^2),$$

where β_0 is the intercept and ϵ is the noise with variance τ^2. Local Solar Time (LST) will be used instead of longitude. It represents the position of a point in Earth's rotation relative to the Sun, where LST 12 is when the Sun is directly above.

4.1 Dataset

Atmospheric mass density data are obtained from HASDM for the Gannon geomagnetic storm period (May 9–13, 2024). Latitude and LST are taken directly from the HASDM grid without additional refinement. The analysis is restricted to an altitude of 400 km to remain consistent with the region of interest used in Exp1 and eliminate inconsistencies within the interpolation, since density decreases exponentially with altitude.

Geomagnetic and solar activity indices (Ap, Dst, and f10.7) are obtained from the OMNIWeb database, which provides a dataset of solar wind and geomagnetic measurements from multiple spacecraft and ground-based sources. These indices are provided at 3-hour resolution and are mapped onto the HASDM grid points in time. After merging, the dataset is treated as a five-dimensional regression problem (excluding time as a feature). The final dataset contains 18,240 samples.

LST is transformed using a periodic representation so the deviation from LST 12 is considered:

$$\text{LST}_\text{p} = \cos\left(\frac{2\pi(\text{LST} - 12)}{24}\right),$$

The regression inputs are then formed as

$$X = [\text{LAT}, \text{LST}_\text{p}, \text{Dst}, \text{Ap}, \text{f10.7}],$$

and normalized using z-score scaling:

$$X_n = \frac{X - \mu_X}{\sigma_X}.$$

The normalization ensures consistent scaling across predictors with differing physical units and magnitudes. The response variable is defined as

$$y = \log(\text{Density}),$$

to reduce dynamic range and improve numerical stability in the regression.

4.2 Priors

The regression coefficients are assigned normal priors reflecting expected physical relationships between atmospheric density and the parameters. The intercept is given a broad normal prior to allow flexibility:

$$\beta_0 \sim \mathcal{N}(0, 16^2).$$

Latitude is assigned a uniform prior over a wide range:

$$\beta_1 \sim \mathcal{U}(-5, 5),$$

allowing the data to determine the strength and sign of its effect.

LST is expected to have a positive effect on density due to solar heating, and is assigned a mildly informative prior:

$$\beta_2 \sim \mathcal{N}(0.5, 1).$$

The Dst index is negatively correlated with geomagnetic activity (more negative values correspond to stronger storms), which are associated with increased density. This motivates a mildly informative prior centered on a negative mean:

$$\beta_3 \sim \mathcal{N}(-1, 1).$$

The Ap index represents geomagnetic activity, with higher values indicating stronger disturbance, and is assigned a positive centered mildly informative prior:

$$\beta_4 \sim \mathcal{N}(1, 1).$$

Similarly, the f10.7 solar flux index is a proxy for solar EUV radiation, which increases thermospheric heating and density, motivating:

$$\beta_5 \sim \mathcal{N}(1, 1).$$

4.3 HMC Regression

The regression is performed using HMC with the following initialization and tuning parameters:

$$n_{\text{iterations}} = 10000, \quad ds = 3 \times 10^{-3}, \quad L = 20,$$

where $n_{\text{iterations}}$ is the total number of iterations, ds is the step size, and L is the number of "leapfrog" steps per iteration. A burn-in period of 2000 samples is discarded to reduce dependence on initial conditions:

$$n_{\text{burn}} = 2000.$$

The observation noise scale is estimated from the data as

$$\tau = \text{std}(y),$$

which is used to characterize the residual variance in the likelihood model.

5. Exp2: Results

The HMC sampler achieved an acceptance rate of 0.8793, indicating stable exploration of the posterior distribution. Trace plots in Figure 12 show the estimation of the coefficients over iterations and indicate a well-mixed chain consistent with convergence. Minor autocorrelation is observed in β_4 and β_5, but overall sampling behavior remains stable.

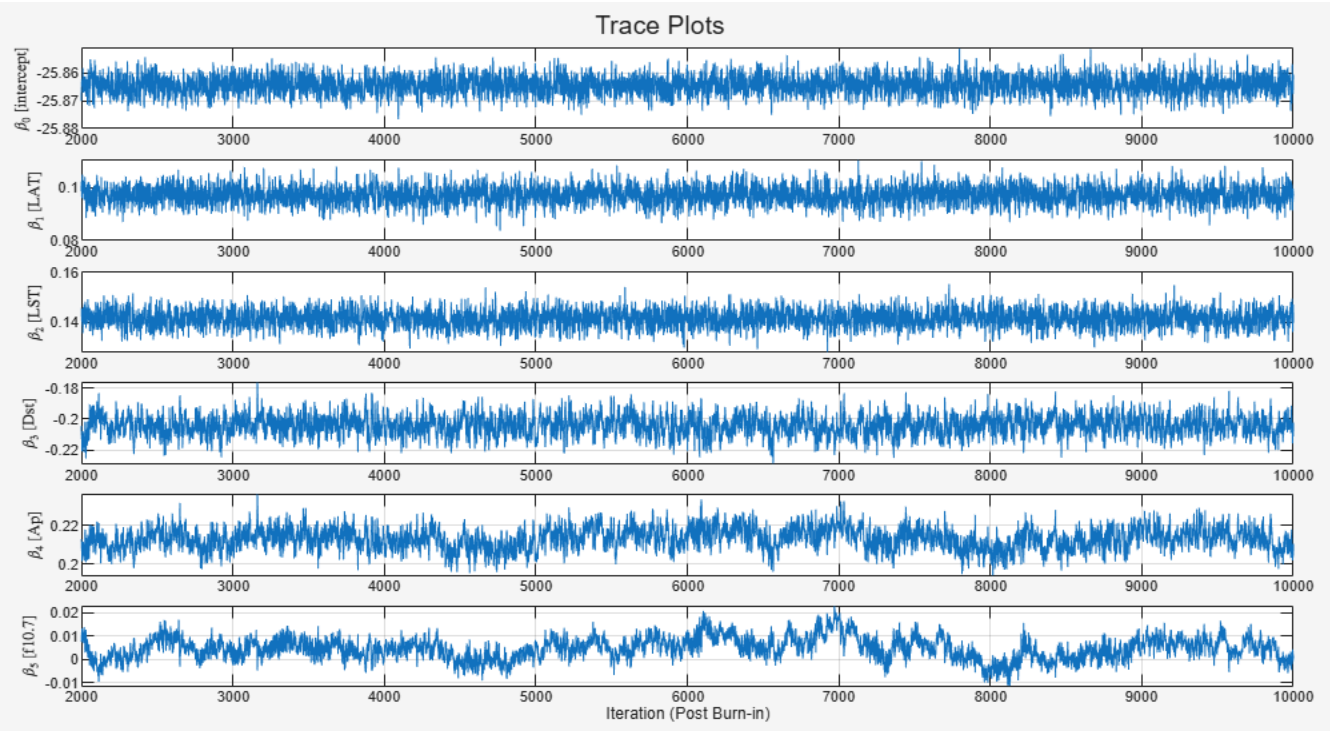

Figure 12: Trace plots of regression coefficients across HMC iterations.

Posterior distributions of the regression coefficients are shown in Figure 13 and are approximately Gaussian. The dashed magenta lines represent the 95% credible intervals and the solid red lines are the means. The intercept β_0 differs significantly in scale from the remaining coefficients, consistent with the magnitude of log(Density), where density values at 400 km are on the order of 10^{-11} to 10^{-12}. The f10.7 coefficient β_5 is closest to 0 and contains 0 in the credible interval, suggesting a relatively weak influence on density compared to geomagnetic indices. This is consistent with previous findings that geomagnetic drivers are stronger predictors of thermospheric density variability than solar flux alone [5].

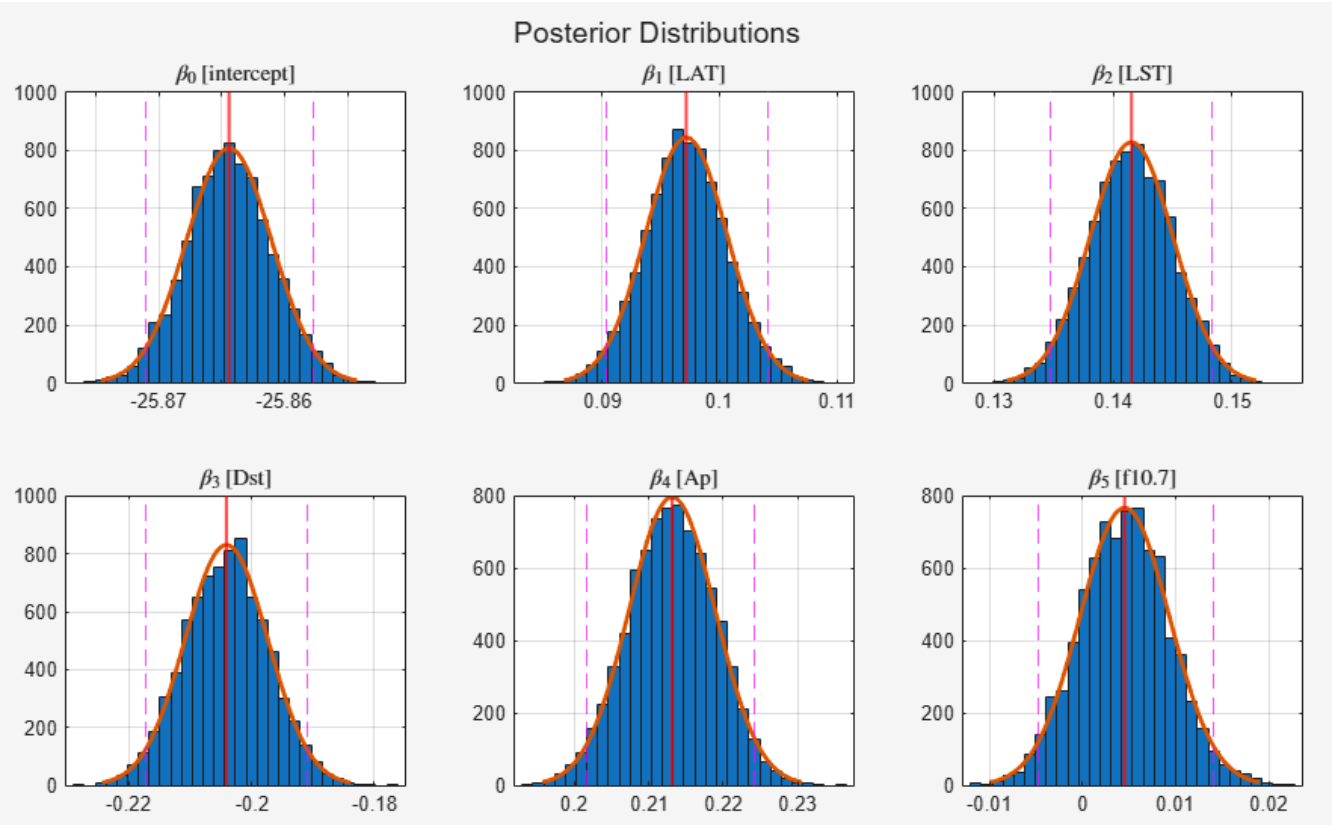

Figure 13: Posterior distributions of regression coefficients.

Posterior mean estimates are summarized in Fig. 14. The Dst coefficient β_3 is negative, consistent with its physical interpretation where more negative values correspond to increased geomagnetic activity and higher density. The f10.7 β_5 coefficient has the smallest magnitude among predictors, indicating a comparatively weaker contribution.

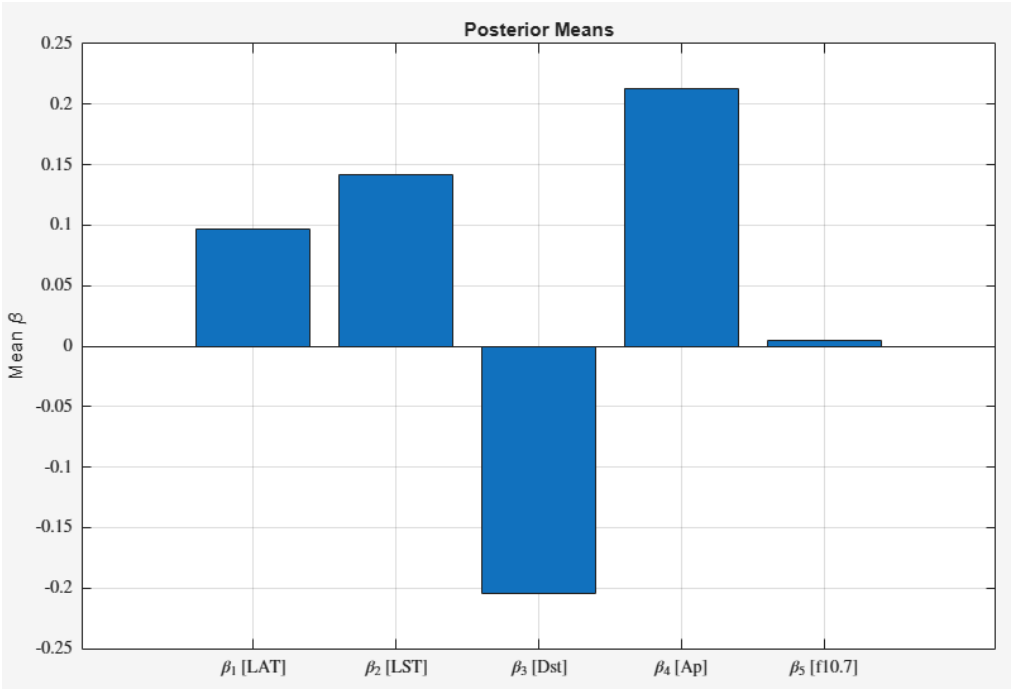

Figure 14: Posterior mean estimates of regression coefficients.

Convergence diagnostics were evaluated using three independent chains initialized with different random seeds. Table 1 summarizes posterior statistics and $\hat{R}$ values. All parameters exhibit $\hat{R} \approx 1.000$, indicating strong convergence. Posterior variances are small relative to coefficient mean magnitudes, further supporting the parameter estimates.

Parameter	Mean	Variance	$CI_{2.5}$	$CI_{97.5}$	$\hat{R}$
Intercept (β_0)	-25.864	1.15×10^{-5}	-25.871	-25.858	0.99999
Latitude (β_1)	0.0972	1.21×10^{-5}	0.0904	0.1042	0.99994
LST (β_2)	0.142	1.23×10^{-5}	0.1348	0.1484	0.99994
Dst (β_3)	-0.204	4.57×10^{-5}	-0.2172	-0.1909	1.0002
Ap (β_4)	0.213	3.37×10^{-5}	0.2017	0.2242	1.0008
f10.7 (β_5)	0.00454	2.33×10^{-5}	-0.00474	0.0142	1.0032

Table 1: Posterior statistics regression parameters.

6. Exp2: Conclusions

The HMC regression provides a physically consistent characterization of atmospheric density. The sampler demonstrates strong convergence, and the posterior distributions are well-behaved, with $\hat{R} \approx 1$ for all parameters and low posterior variance relative to coefficient magnitude. It is important to note that the dataset used in this analysis is model-derived (HASDM), even though it considers real observational data. As a result, performing regression on this output may partially invert the structure of the original physical model.

Results indicate that geomagnetic indices (Dst and Ap) and LST have the strongest and most consistent influence on density, with signs and magnitudes consistent with expected response to geomagnetic activity. In contrast, the f10.7 solar flux term exhibits a weak effect, suggesting that, for this dataset and storm-time interval, solar forcing is less directly predictive than geomagnetic forcing. This provides a statistical foundation for linking space weather indices to density fields and supports their use in improved density modeling for orbit uncertainty applications.

6.1 Future Work

Future work can extend both the temporal and spatial scope of the regression framework. Increasing the time range beyond the current storm interval and incorporating longer-term datasets would enable assessment of model robustness across different space weather regimes. Expanding the analysis across multiple altitudes would further allow direct comparison of density drivers in different thermospheric regimes.

Additional predictors can also be incorporated to improve physical fidelity, including expanded geomagnetic and solar wind parameters such as solar wind speed, interplanetary magnetic field strength, and additional geomagnetic indices. This would provide a more complete representation of external forcing on atmospheric density variability.

Finally, the regression model can be validated by reconstructing density fields from posterior parameter estimates and comparing predicted density against HASDM observations. This would provide a direct assessment of predictive performance and quantify the practical utility of the inferred model for density estimation and orbit uncertainty applications.

Appendix: Code Access

All analysis for this project was implemented in MATLAB. The code is available on GitHub and Google Drive:
`https://github.com/emmettpet-95/STAT5630Project`.
`https://drive.google.com/file/d/1Ug7FEg0YqIsXA3vjCrijKoWBS6y4e_or/view?usp=sharing`

The `Function` and `Analysis` directories contain functions, while the `Plots` directory includes plotting utilities. The `Saved Data` directory stores datasets and results to avoid re-running computationally intensive scripts. These data take up over 1.5 GB and were too large to upload to GitHub.

For Exp1, the main scripts are `A_PastResults_alpha`, `B_GridSetup`, `C_Mean_and_NoDrag_Runs`, and `D_Propagate_Orbits`. The script `A_PastResults_alpha` imports prior results and defines the α distributions. `B_GridSetup` loads HASDM data and constructs the working grid. `C_Mean_and_NoDrag_Runs` performs baseline orbit propagations, and `D_Propagate_Orbits` executes Monte Carlo simulations for the different density cases and generates results.

For Exp2, the scripts `A2_GridSetup` and `B2_Regression` are used. `A2_GridSetup` prepares the dataset and regression inputs, while `B2_Regression` performs the HMC regression and produces the associated results and plots.

References

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